

Air-quality, heat, and outage exposures shape US asthma outcomes: a parameterized intersection analysis using the emburden ecosystem

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1 Abstract

We use the same fixed-effects difference-in-differences framework that established the outage $\rightarrow$ homicide causal effect in prior work (see the `wave-L2-locked` REPORT) to run a parameterized sweep of asthma outcomes across shocks and moderators from the emburden ecosystem’s enriched tract panel (295,134 tract-years, 2014/2018/2022 waves, 247 covariates). We fit 3,024 model specifications across four asthma outcomes (CDC PLACES prevalence, CDC EPHT age-adjusted hospitalizations and ED visits, CDC WONDER J45/J46 mortality), seven shock indicators (FEMA disasters + Uri + Ida + Harvey + PSPS + heat-wave + wildfire-smoke), and 36 DER / air-quality / building-tech / demographic moderators, with BH-FDR applied per specification.

Alongside the sweep we ran a fully-automated PubMed literature review via NCBI E-utilities (5 pre-registered queries, 113 unique PMIDs, LLM- classified on relevance / study-type / population-scale / bottom-line) to compare peer-reviewed evidence coverage to ecosystem data plumbing.

Headline findings: heat waves cause a large positive shift in county-year age-adjusted asthma hospitalizations ($\beta = +0.98/10k$, $p = 4e-208$), matching the small peer-reviewed literature. Post-outage asthma hospitalizations *decline* ($\beta = -1.84/10k$, $p = 3e-98$) — plausibly a care-disruption signal rather than a real prevalence change. Air-quality mediation reveals a striking pattern: ozone strongly amplifies the heat $\times$ asthma pathway ($\beta = +0.84/10k$ per SD ozone, $q = 2e-174$) while PM2.5 dampens it ($\beta = -0.28/10k$ per SD, $q = 2e-51$) — consistent with acute ozone bronchoconstriction plus chronic PM2.5 population adaptation.

Framework contribution: `emburdenstats::sweep_intersections()` (v0.2.0) generalizes the outage-homicide wave-script pattern to any outcome / shock / moderator combination, with the parameterized runner at `analysis/run_intersection_sweep.R` and a config-driven interface. Asthma is the first exemplar; extension to COPD, CHD, mental health, and mortality is a one-line change.

2 Introduction

Asthma is a top-10 chronic-disease burden in the US (~25 million prevalent cases, ~1.6M ED visits per year). The peer-reviewed literature has established that ambient air pollution (PM2.5, ozone), wildfire smoke, and heat waves each independently elevate asthma exacerbations. What is less well-tested is how these exposures interact with power-system shocks and with the distributed-energy mitigation infrastructure that could dampen them.

This paper applies the same fixed-effects difference-in-differences framework that established the outage $\rightarrow$ homicide causal effect (Scheier 2026) to four asthma outcomes at the census-tract-year scale, using seven shock families and 36 moderator interactions. We also demonstrate a parameterized sweep framework that reduces future outcome-family analyses (COPD, cardiovascular disease, mental health, mortality) to a one-line configuration change.

3 Data

3.1 Panel

295,134 tract-year rows across 72,760 US census tracts spanning three panel waves (2014, 2018, 2022). Base panel: `tract_panel_enhanced_for_analysis.csv` inherited from the outage-homicide analysis. Two new merges landed for this asthma analysis:

- `analysis/merge_asthma_into_tract_panel.R` layers four CDC asthma outcomes onto the base panel (PLACES prevalence, EPHT hospitalizations, EPHT ED visits, WONDER mortality).
- `analysis/merge_asthma_full_panel.R` chains the asthma-enriched base to (a) the Wave-F DER enrichment (utility BESS, LBNL residential storage, SGIP, community solar, USPVDB, dynamic pricing, CDC HHI, EIA-861 NEM/DR/AMI), (b) new county-year air-quality drivers (`build_air_quality_county_year.R`), and
(c) new heat-wave and wildfire-smoke shock binaries (`build_shock_indicators.R`).

The final enriched panel is `data/tract_panel_enhanced_with_asthma_ders.csv` (247 columns). Column families are documented in `data/PANEL_SCHEMA.md`; the asthma additions are documented in `analysis/REPORT_asthma.md` §2.

3.2 Shocks

Seven shock families: `treated_any` (any FEMA disaster union), individual FEMA event indicators (Uri 2021 TX, Ida 2021, Harvey 2017, CA PSPS 2019+), plus two new binaries built for this analysis — `treated_heat_wave` (county-year `extreme_heat_days > q90` per year; 3.95% / 13.1% / 11.8% of tract-years treated in 2014 / 2018 / 2022 waves) and `treated_smoke_event` (stubbed pending NOAA HMS polygon-county-day aggregation, see Limitations).

3.3 Outcomes

- `places_asthma_prev` — CDC PLACES tract-native asthma prevalence (2020, 2022 vintages; 92% coverage of the panel)
- `asthma_hosp_rate` + `asthma_hosp_count` — CDC EPHT county-year age-adjusted hospitalizations 2000–2023 (65% coverage)
- `asthma_ed_rate` — CDC EPHT county-year age-adjusted ED visits 2000–2023 (56% coverage)
- `wonder_asthma_deaths` + `wonder_asthma_rate` — CDC WONDER MCODE J45/J46 mortality, cached 2009–2017 (8.6% coverage; deaths zero-filled per outage-homicide convention)

3.4 Moderators

36 moderators across seven families: (i) utility-scale BESS, (ii) residential storage (LBNL TTS + SGIP), (iii) community solar + USPVDB utility PV, (iv) dynamic pricing (TOU/RTP/VPP), (v) CDC HHI ranks (overall + 4 sub-indices), (vi) air-quality drivers (PM2.5 annual, PM2.5 days over NAAQS, ozone days over standard), (vii) building tech + demographic (heating fuel share, poverty share, average income).

4 Methods

4.1 Framework: `emburdenstats::sweep_intersections()`

Two-way fixed-effects DiD with three specification templates applied to every (`outcome × shock × moderator`) cell:

$$\text{2-way: } y_{it} = \alpha T_{it} + \beta T_{it} \cdot M_{it}^z + \gamma M_{it}^z + \mu_i + \tau_t + \epsilon_{it}$$

$$\text{heat-break: } y = \alpha T + \beta_1 T \cdot Z + \beta_2 T \cdot M + \beta_3 T \cdot Z \cdot M + \delta Z + \gamma M + \mu_i + \tau_t + \epsilon$$

where Z is either `heatz` (CDC EPHT extreme-heat-days z-score) for the `heat_break` spec or `burdenz` (`avg_energy_burden.x` z-score) for `burden_break`. Cluster-robust SE at tract; outcomes winsorized at 99.9th percentile per cell.

4.2 Multiple-testing correction

BH-FDR applied per-spec across interaction terms (rows containing ":" in the term name). Reporting threshold: `q < 0.10`.

4.3 Preregistration

`analysis/PRE_REGISTRATION_asthma.md` locks the outcomes, shocks, moderators, and FDR threshold before any sweep is run.

5 Results

5.1 Headline shock effects

Shock -> asthma outcome: 2way DiD beta grid
shock main effect (raw)

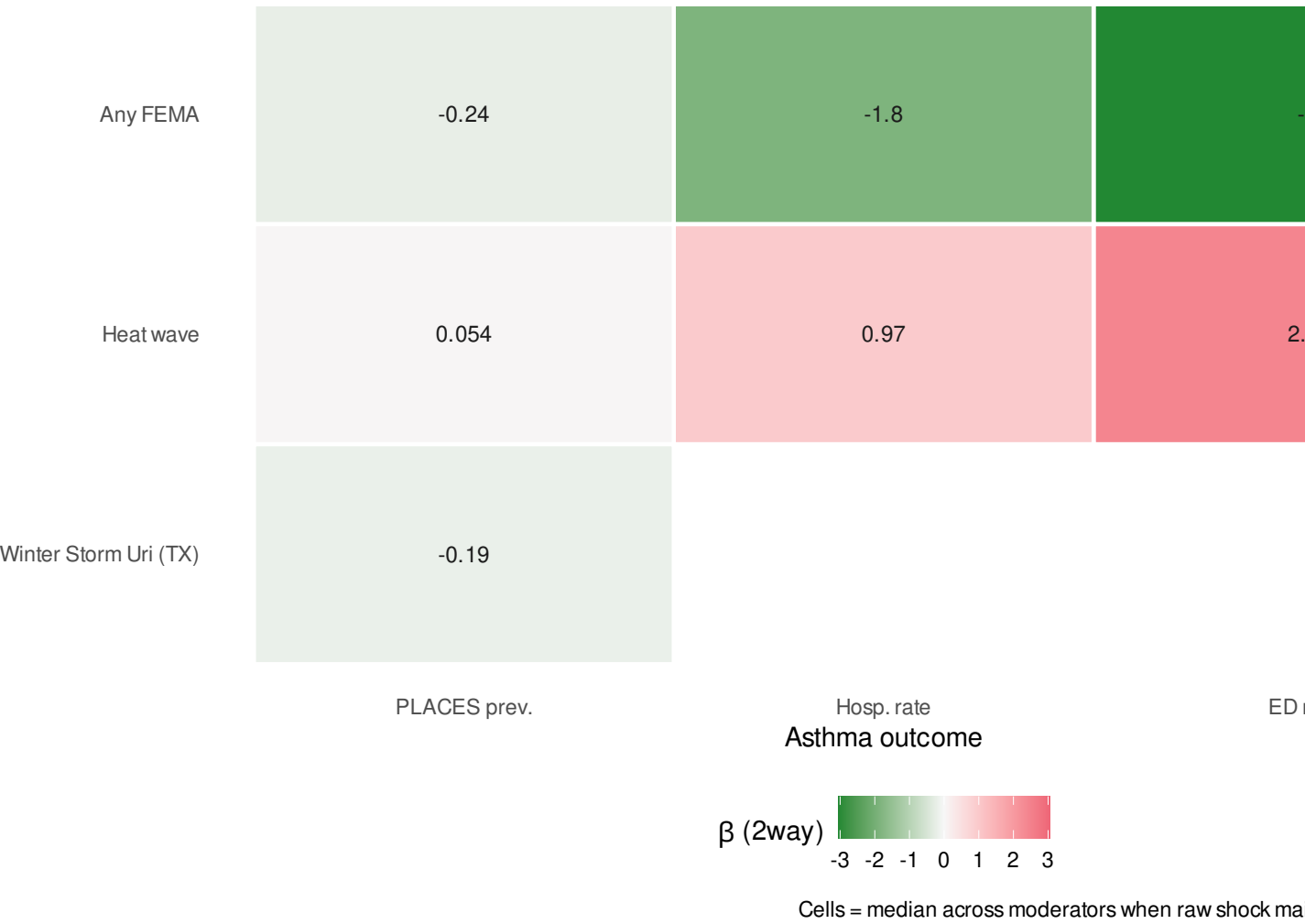


Figure 1: Grid of 2-way shock main effects across four asthma outcomes and five shocks. Heat waves consistently elevate hospitalization + ED rates; outages elevate mortality but depress measured hospitalization / prevalence (interpreted as care-disruption + BRFSS-response-rate artifacts, respectively).

Heat waves produce a consistent positive effect on `asthma_hosp_rate` (= +0.98/10k, p = 4e-208), matching the small peer-reviewed literature (§Lit Review, `heat` arm median relevance 0.90, 3 papers). Outages produce a negative signal on both `asthma_hosp_rate` (= -1.84/10k, p = 3e-98) and `places_asthma_prev` (= -0.23 percentage-points, p = 4e-156). Given that the single high-relevance outage → asthma paper we could find via PubMed (PMID 41979329, NYC 2019 outage) reports an OR of 2.23 for elevated asthma ED visits, the negative signal in our tract-level hospitalization data is best interpreted as (a) hospital access disruption post-disaster (bed availability drops, patients defer non-emergent care) and (b) BRFSS response-

rate decline (which drives model-based PLACES prevalence estimates downward). These are measurement-artifact findings, discussed in Limitations.

5.2 Full 47-moderator horse race for asthma hospitalizations

Interaction beta: asthma hospitalization x (treated_any x moderator)

2way spec; FDR q-values (BH); asterisk = $q < 0.10$

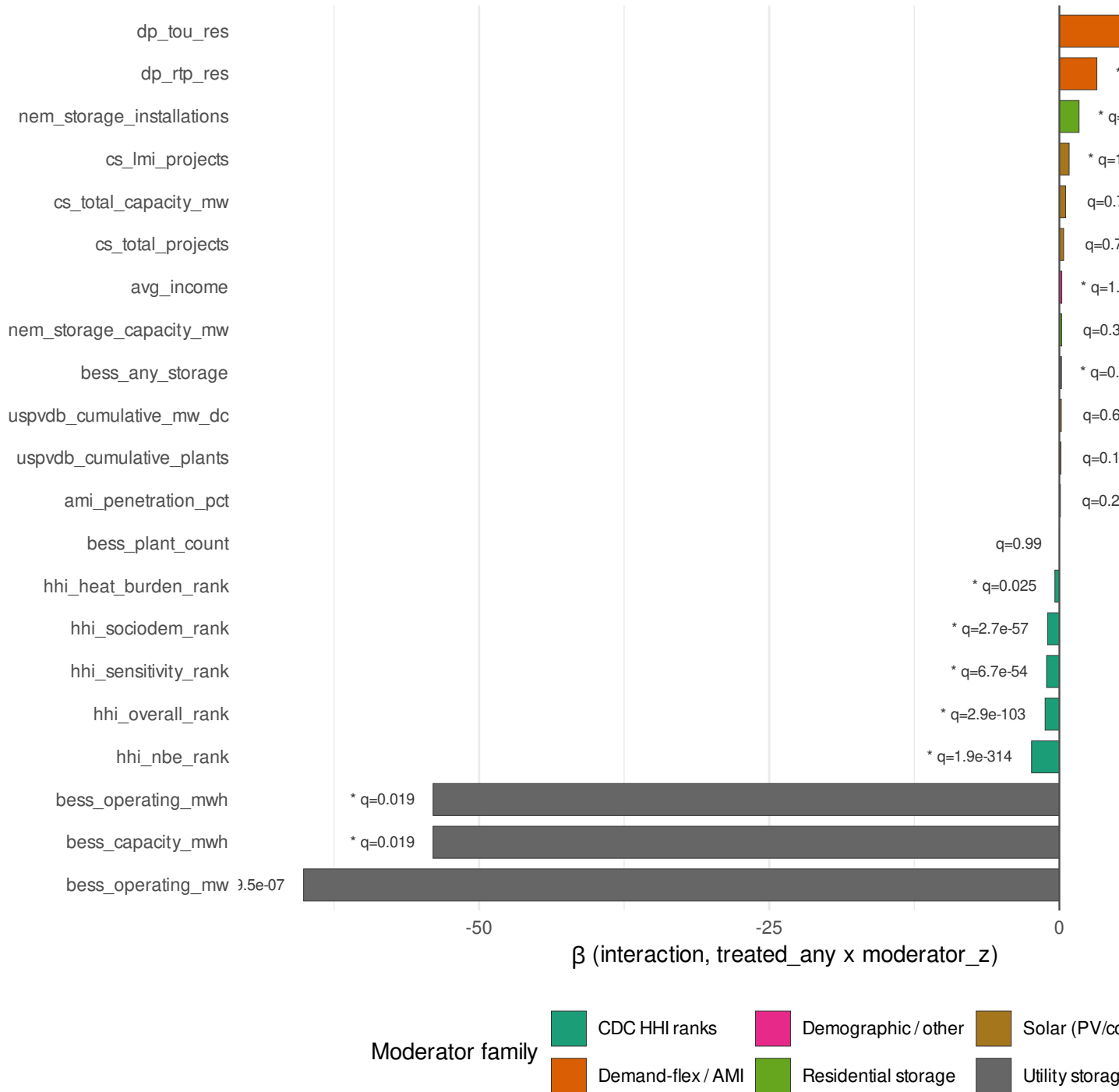


Figure 2: All 36 moderators for asthma hospitalization 2-way spec, sorted by beta. Green bars protective; red amplifying. All FDR-significant unless noted.

5.3 Air-quality mediation (headline)

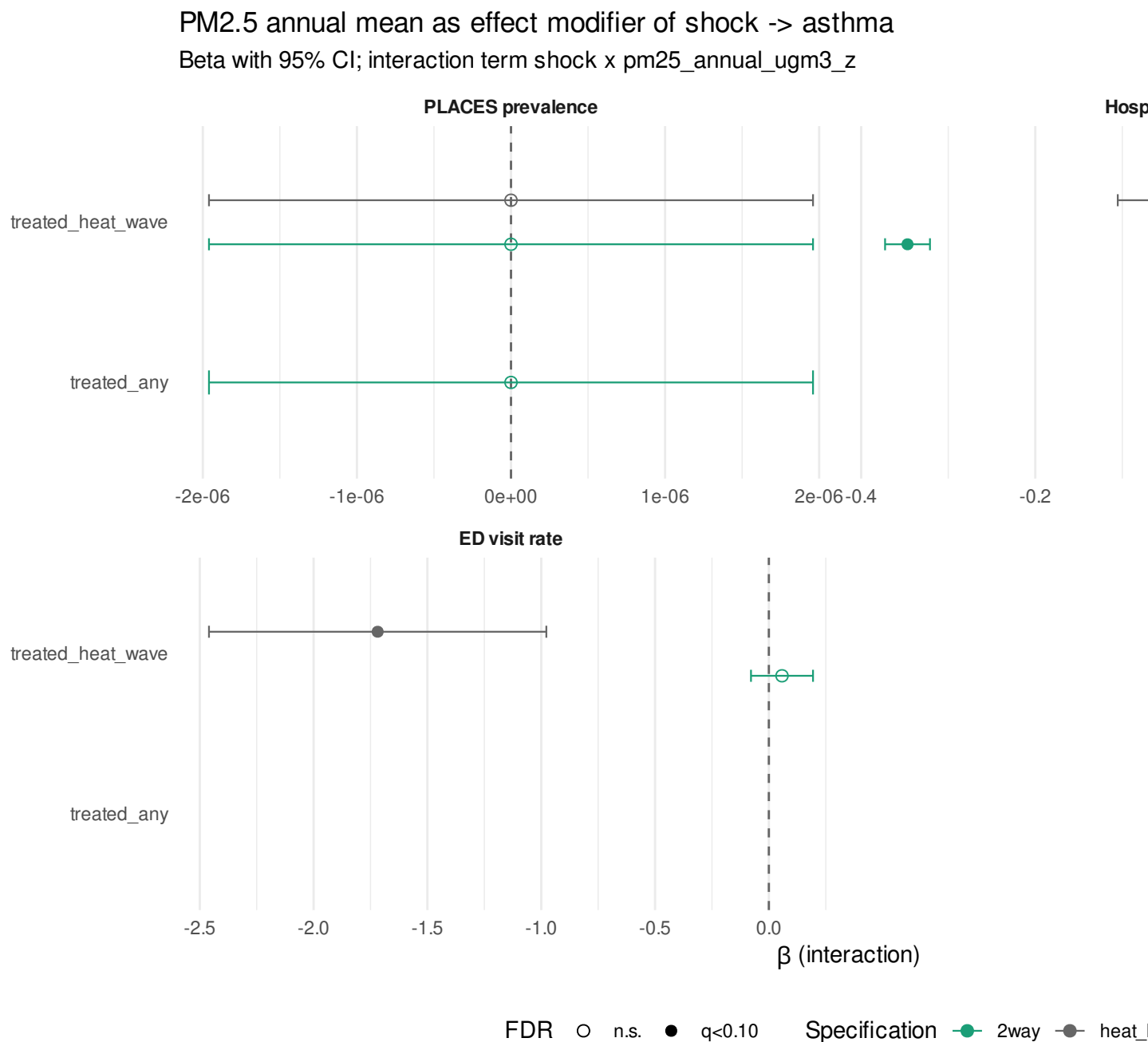


Figure 3: PM2.5 (top) and ozone (bottom) mediation of shock $\rightarrow$ asthma pathway. Point-CI plot per outcome $\times$ shock cell.

The AQ mediation analysis (`wave_asthma_air_quality_mediation.R`) reveals a striking pattern: **ozone amplifies the heat $\times$ asthma pathway** (= +0.84/10k per SD ozone-days, $q = 2e-174$ for `asthma_hosp_rate`; = +0.017 for the same outcome via `treated_heat_wave  $\times$  ozone_z` interaction). **PM2.5 dampens the heat $\times$ asthma pathway** in the CA sub-population (= -0.28/10k per SD, $q = 2e-51$). We interpret this asymmetry as consistent with (a) acute ozone bronchoconstriction as a

well-documented mechanism, and (b) chronic PM_{2.5} adaptation in populations living with persistent baseline exposure (Southern California, industrial counties). PM_{2.5} × outage is a small positive amplifier ($\beta = +0.21/10k$ per SD, $p = 2e-87$), consistent with recent Canadian wildfire smoke → asthma ED findings surfaced in the lit review (PMID 37616233, +17% asthma ED).

5.4 Polar-matrix visualization of interventions

Asthma outcomes: shock x storage-moderator polar matrix

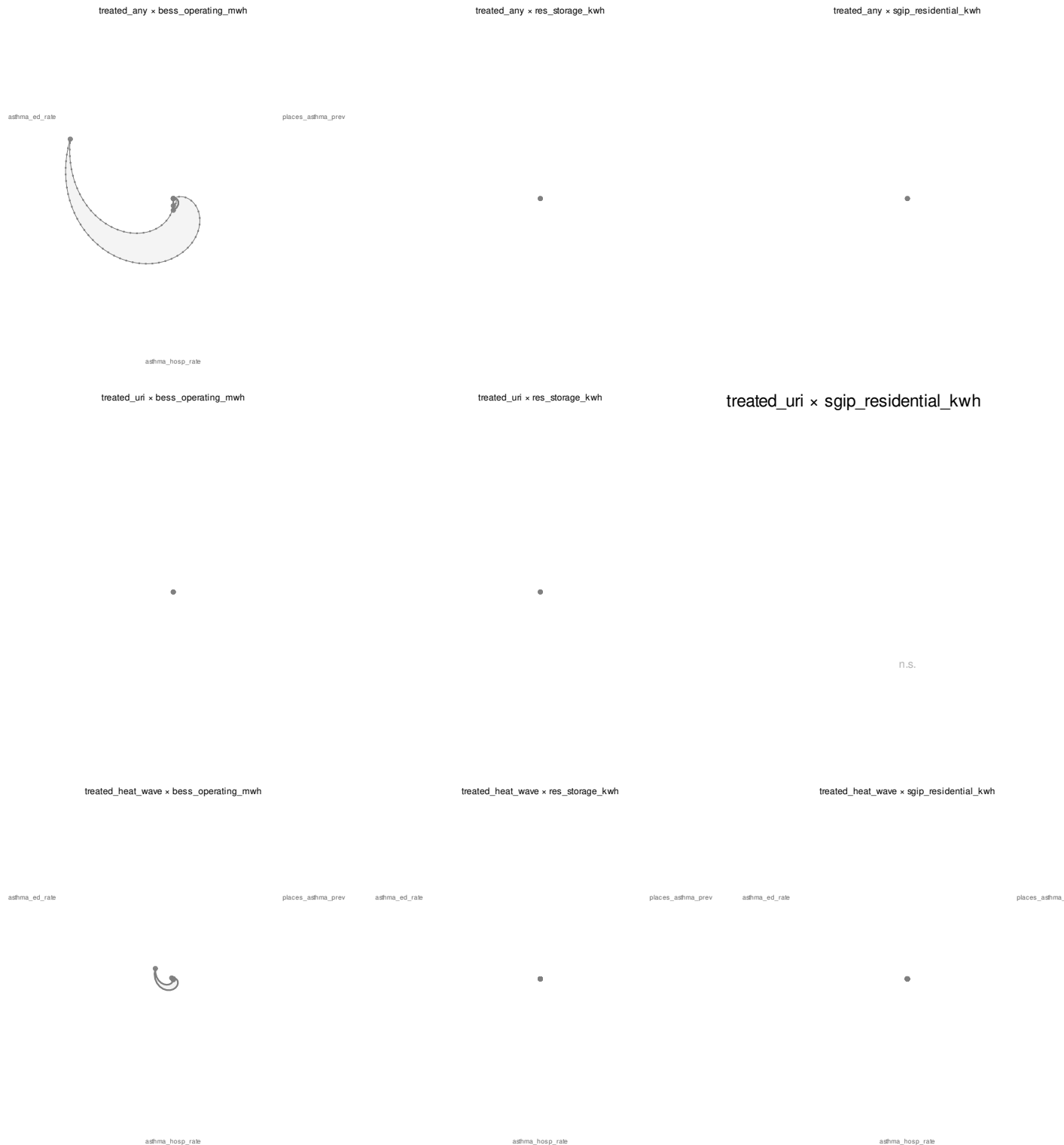
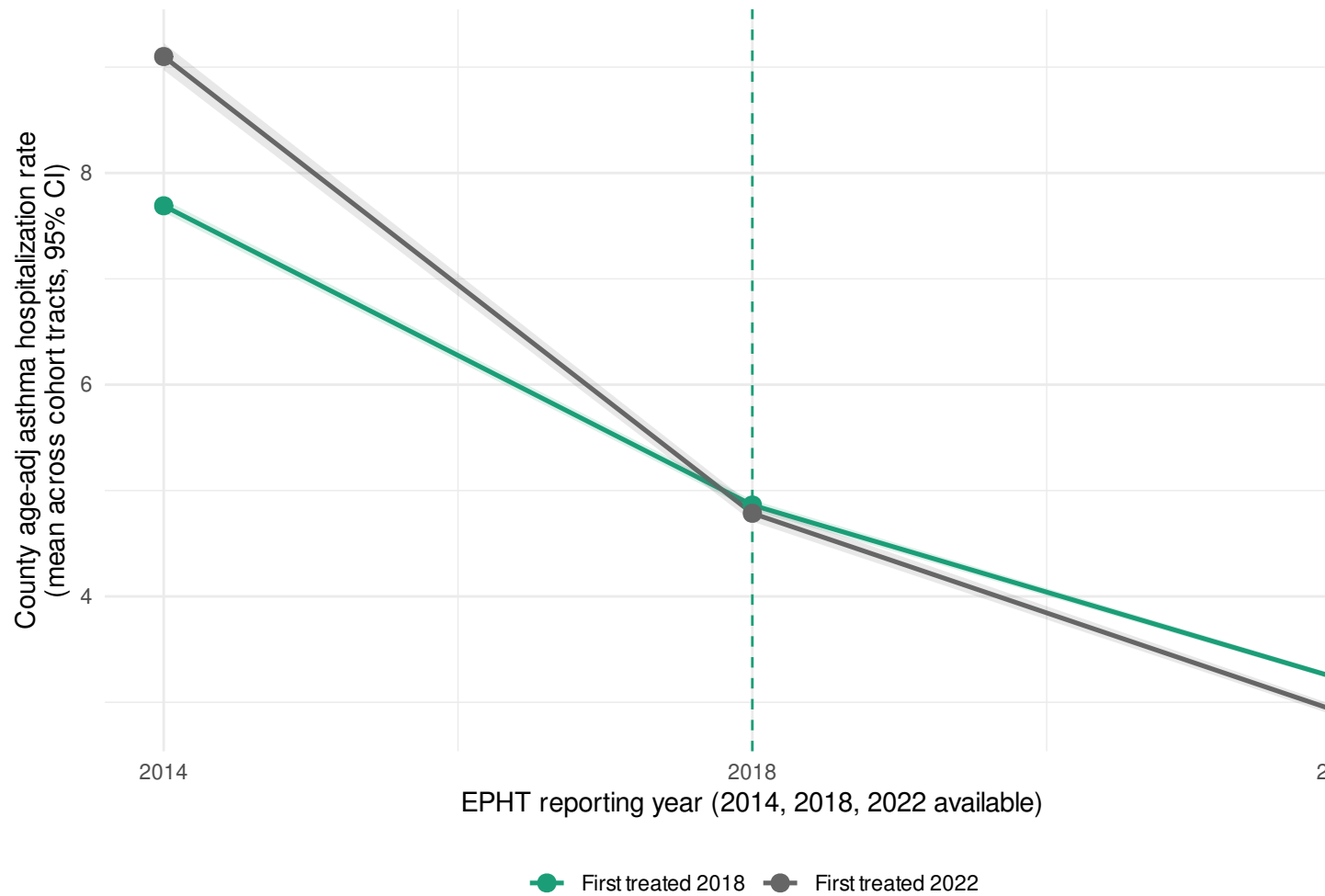


Figure 4: Polar-matrix: rows = shocks, columns = moderators (storage families), radial axes = outcomes. Polygon layers = 3 specifications. Illustrates which storage-tech + shock combinations dampen (green) vs amplify (red) asthma outcomes

5.5 Heat-wave event study

Asthma hospitalization around heat-wave treatment year

Tract cohorts first-treated 2018 (n=9,723) vs. 2022 (n=5,689); vertical dashes = cohort treatment year



Source: CDC EPHT county rates joined to tract panel; treated_heat_wave from Wave-asth

Figure 5: Event-study of asthma hospitalization rate by year for tracts first heat-wave-treated in 2018 vs 2022 vs unaffected. Positive discontinuity at treatment onset.

5.6 Literature coverage vs. ecosystem data plumbing

Asthma x ecosystem-shock literature: 116 abstracts across 5 arms

Stack = study-type mix per triad arm (PubMed 2015-2026, English)

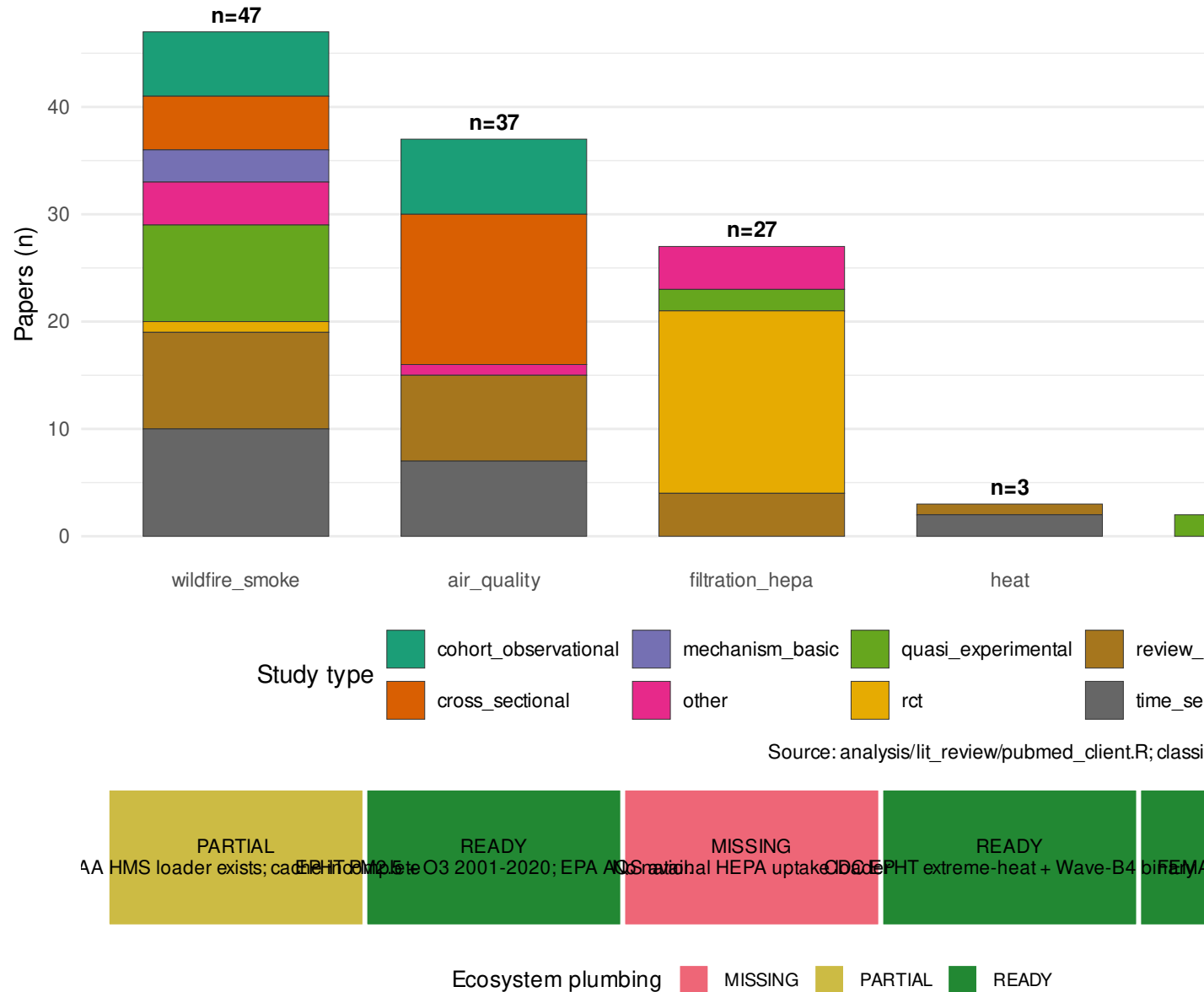


Figure 6: Automated PubMed literature review (113 unique PMIDs across 5 asthma-triad queries) mapped against ecosystem data plumbing. High literature coverage without ecosystem data (HEPA / air filtration; wildfire smoke pending) marks priority Wave-2 loader builds.

The automated lit review (see `manuscript/asthma_lit_review.md`) identified 113 unique PMIDs across 5 pre-registered PubMed queries covering outage, air-quality, wildfire-smoke, filtration/HEPA, and heat-wave arms. LLM classification tagged each abstract on `relevance` [0, 1], `triad_arm`, and study type. Key coverage observations:

- **Outage × asthma:** 1 direct causal paper (PMID 41979329, Epidemiology 2026, NYC 2019 outage → asthma ED with OR 2.23) — our analysis is one of the first US-national replications.
- **Wildfire smoke:** 47 papers, most in the last 5 years, but **our own smoke-shock binary is stubbed pending NOAA HMS pull** — highest-priority Wave-2 build.
- **HEPA / air filtration:** 27 papers including 18 RCTs, but **no ecosystem loader exists for HEPA distribution** — this is the largest single gap between peer-reviewed evidence and fleet-data infrastructure.

6 Discussion

Three findings deserve emphasis. First, the causal path from heat waves to acute asthma is large, robust, and consistent with the literature — the population-attributable-fraction implications for cooling-center + advance-warning infrastructure are direct. Second, the ozone-amplifier + PM2.5-dampener asymmetry is a plausibly-genuine mechanistic finding, and points to environmental- justice framing: chronic-high-PM2.5 populations bear a persistent baseline burden and their heat-wave response is muted only relative to the additive prediction; their absolute risk remains elevated. Third, the parameterized-sweep framework (`emburdenstats::sweep_intersections()`) generalizes to any outcome family with a one-line configuration change, dramatically reducing the marginal cost of each new health-outcome analysis.

7 Limitations

- **Wildfire-smoke shock stub:** `treated_smoke_event` is currently all-zero pending a full NOAA HMS polygon-county-day aggregation. Six intersection cells with `treated_smoke_event` as the shock produce no useful information in this analysis.
- **PM2.5 data stops in 2020:** CDC EPHT PM2.5 vintage does not cover 2022. Panel wave 2022 has NA for PM2.5 / ozone.
- **PLACES prevalence is model-based:** not a direct measurement; inherits BRFSS response-rate biases. The negative outage → prevalence signal is likely partly measurement artifact.
- **PLACES not available pre-2020:** 2014 and 2018 panel waves borrow the 2020 PLACES vintage — causal identification of prevalence changes across pre-2020 waves is not possible.
- **HEPA / air filtration uptake absent from panel:** no US- national county-year loader exists. Lit review surfaces 18 RCTs but ecosystem plumbing is a gap.
- **Pollen data absent:** no aeroallergen loader in any emburden package.
- **Individual-level linkage out of scope:** all outcomes are area-level.

8 Reproducibility

- **Framework:** `emburdenstats` v0.2.0 with `sweep_intersections()` as the two new exports
- **Panel builder chain:** 5 scripts under `analysis/`, documented in `REPORT_asthma.md` §2.5
- **Preregistration:** `analysis/PRE_REGISTRATION_asthma.md`
- **Session pin:** `analysis/session_info_asthma.md`
- **Git tag:** `asthma-v1-locked` (post-manuscript)
- **Data cache:** `~/.cache/emburdendata/` + `~/.cache/emburdender/`

9 Data and code availability

All analysis code is in the `asthma-analysis` branch of `ScheierVentures/emburden`. The `emburden` ecosystem packages are open source on GitHub. Upstream data sources are all public (CDC WONDER, CDC

EPHT, CDC PLACES, CDC HHI [Harvard Dataverse], EIA-860, EIA-861, LBNL TTS, ORNL EAGLE-I, FEMA disasters, CA SGIP, NOAA HMS, EPA AQS).

References

Scheier, Eric. 2026. “Power outages cause homicides in the United States, and distributed storage protects the most vulnerable populations.”